| Program : **Diploma in Architecture** | |
|---|---|
| Course Code : **3182** | Course Title: **Climate responsive architecture** |
| Semester : **3** | Credits: **3** |
| Course Category: **Program Core** | |
| Periods per week: **3 (L:2, T:1, P:0)** | Periods per semester: **45** |

**Course Objectives:**

- To identify the influence of tropical climates on architecture and apply it on the design of shelters for human comfort.

**Course Prerequisites:**

| Topic | Course code | Course Title | Semester |
|---|---|---|---|
| Global climatic factors | | Basic Geography | Secondary |
| Modes of heat transfer | | Applied Physics I | I |

**Course Outcomes:**

On completion of the course, the student will be able to:

| COn | Description | Duration (Hours) | Cognitive Level |
|---|---|---|---|
| CO1 | Identify climatic elements, characteristics of different types of tropical climates and factors affecting site climate. | 10 | Understanding |
| CO2 | Explain human thermal comfort factors and summarize how structural controls affect thermal design of buildings in order to attain human comfort | 11 | Understanding |
| CO3 | Explain the effect of ventilation and air movement and other passive design techniques in cooling buildings | 11 | Understanding |
| CO4 | Summarize design features of shelters in tropical climates | 11 | Understanding |
| | Series Test | 2 | |

**CO-PO Mapping:**

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|---|---|---|---|---|---|---|---|
| **CO1** | 3 | | | | | | |
| **CO2** | 3 | | | | | | |
| **CO3** | 3 | | | | 2 | 2 | 2 |
| **CO4** | 3 | | | | | | |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline:**

| Module Outcomes | Description | Duration (Hours) | Cognitive level |
|---|---|---|---|
| **CO1** | **Identify climatic elements, characteristics of different types of tropical climates and factors affecting site climate.** | | |
| M1.01 | Explain principal climatic elements | 2 | Understanding |
| M1.02 | Classify tropical climates and their characteristics | 4 | Understanding |
| M1.03 | Summarize factors affecting site climate and urban climate | 4 | Understanding |
| **Contents:**<br>Climate and weather - definitions; Principal climatic elements - Solar radiation- quality, solar constant; air temperature - measurement, data; humidity - measurement; wind - Trade winds - thermal forces, Westerlies and Polar winds, global wind pattern (figure); precipitation - definition, measurement, driving rain - Tilt of the earth's axis - Earth's thermal balance - Sky conditions, vegetation<br>Classification of tropical climates - Tropical climates - Warm humid climate - Warm humid island climate - Hot dry desert climate - Hot dry maritime desert climate - Composite or monsoon climate - Tropical upland climate - characteristics of all these climates.<br>Site climate - Definitions of macro and micro climates and site climate, local factors affecting site climate - topography, ground surface, three dimensional objects (Brief explanation only) Urban climate - factors causing deviation of urban climate from regional macroclimate - changed surface qualities, buildings, energy seepage, atmospheric pollution. | | | |
| **CO2** | **Explain human thermal comfort factors and summarize how structural controls affect thermal design of buildings in order to attain human comfort** | | |
| M2.01 | List thermal comfort factors and summarize regulatory mechanisms of human body | 2 | Understanding |
| M2.02 | Explain heat exchange process of buildings | 2 | Understanding |

| | | | |
|---|---|---|---|
| M2.03 | Identify the effect of structural controls on thermal design of buildings | 7 | Understanding |
| | Series Test - I | 1 | |
| **Contents:**<br>Body's heat production - heat loss - Regulatory mechanisms - Subjective variables. Heat exchange processes of buildings - thermal balance equation, thermal design. Thermal controls - structural controls - need for structural controls - thermal capacity - solar control - greenhouse effect - reduction of solar heat gain through windows - orientation, internal blinds and curtains, heat absorbing glasses, other special glasses, external shading devices - sun's positions-angle of incidence - shadow angles - shading devices - design of shading devices. | | | |
| **CO3** | **Explain the effect of ventilation and air movement and other passive design techniques in cooling buildings** | | |
| M3.01 | Explain the effect of ventilation and air movement on thermal design of buildings | 6 | Understanding |
| M3.02 | Identify passive cooling techniques in buildings | 3 | Understanding |
| M3.03 | Identify passive cooling methods adopted in a traditional building and make a report after conducting literature study | 2 | Applying |
| **Contents:**<br>Ventilation and air movement - functions of ventilation - supply of fresh air, convective cooling - provision for ventilation - stack effect - physiological cooling-provision for air movement - wind effects - air flow through buildings - orientation, cross ventilation, size of openings, controls of openings - air movement and rain - air flow around buildings-humidity control-working of wind - scoop.<br>Passive design - Passive cooling – orientation - ventilation - convective cooling - air movement - physiological and evaporative cooling, - shading - shading devices, trees & vegetation, shading of roof - thermal insulation - filler slabs, composite walls, cavity walls and double roof - thermal mass - Envelope design - green roof - solar chimney - diode roof - roof pond - earth coupling - earth air tunnel. | | | |
| **CO4** | **Summarize design features of shelters in tropical climates** | | |
| M4.01 | Outline design features of shelters suitable for hot dry climate | 3 | Understanding |
| M4.02 | Summarize design features of shelters suitable for warm humid climate | 3 | Understanding |
| M4.03 | Summarize design features of shelters suitable for composite climate | 3 | Understanding |
| M4.04 | Summarize design features of shelters suitable for tropical upland climate | 2 | Understanding |
| | Series Test -II | 1 | |

| Contents: |
|---|
| Shelter for hot - dry climate-nature of the climate - physiological objectives - form and planning - external spaces - roofs, walls and openings - roof and wall surfaces - ventilation and air flow - traditional shelter.<br>Shelter for warm humid climates - nature of the climate - physiological objectives - form and planning external spaces - roofs and walls - air flow and openings - ventilation - traditional shelter.<br>Shelter for composite climates - nature of the climate - physiological objectives - design criteria - form and planning external spaces - roofs and walls - surface treatment - openings -ventilation and condensation - traditional shelter.<br>Shelter for tropical upland climate - nature of climate - physiological objectives - form and planning - external spaces - roofs and walls l surface treatment - openings - traditional shelter. |

**Text / Reference:**

| T/R | Book Title/Author |
|---|---|
| T1 | Koenigsberger, Ingersoll, Mayhew and Szokolay - MANUAL OF TROPICAL HOUSING AND BUILDING |
| R2 | Givoni - Man, climate and architecture |

**Online resources:**

| Sl. No | Website Link |
|---|---|
| 1 | https://www.researchgate.net/publication/312432251_An_overview_of_passive_cooling_techniques_in_buildings_Design_concepts_and_architectural_interventions |
| 2 | http://www.yourhome.gov.au/passive-design/passive-cooling |
| 3 | https://www.greenroofplan.com/intensive-vs-extensive-green-roofs |
| 4 | https://en.wikipedia.org/wiki/Passive_cooling |
| 5 | https://en.wikipedia.org/wiki/Solar_chimney |